

Appendix How to Write a Report

It has been mentioned that a good report is one where the narrative is complete enough for the reader to picture the essentials of the original record. As with most neurological findings, it is best to describe what you see in simple terms rather than use convoluted terminologies and equivocations. The report will usually fall into the hands of a treating physician who may not be a neurologist. Therefore, it becomes important to avoid EEG related jargon and if this is unavoidable, clearly explain the implications.

It is recommended that the report consists of three main parts:

1. introduction
2. description
3. interpretation

Introduction

This part should include the patient's name, demographics, and identifiers. It should briefly describe the reason for testing and prior preparations such as sleep deprivation, and it should list any relevant medications that may affect the EEG. This should be followed by a brief technical summary describing the system of electrode placement (e.g., standard 10–20 electrodes), any additional modifications, monitoring of other physiological parameters, and the commercial system in use. The total recording time may also be mentioned here.

Description

This is the main body text of the report. It should consist of objective descriptions of the predominant background and overlying foreground patterns, irrespective of whether they are normal or abnormal. Each physiological state during the recording should be described separately. Start with the description of the record during wakefulness followed by drowsiness and sleep. If a certain state was not achieved, then that should be mentioned. This should be followed by a description of the results of activation procedures such as hyperventilation and photic stimulation. If an event (such as a seizure) occurs during the recording, note the time of occurrence (or push button) and approximate duration. A concise description of the event should include the clinical semiology (either witnessed at bedside or on video), accompanying electrographic changes and heart rhythm on single EKG channel. The presence of significant artifact or EKG abnormalities should also be included. Avoid passing judgments about clinical significance in this part of the report.

Interpretation

This should consist of the reader's *impression*, that is, a subjective statement if the recording is normal or abnormal with relevant reasoning. If abnormalities occur, readers should further comment if they are epileptic or ictal. The record should also be compared to any prior studies, if these are available for review. Separately, a *clinical correlation* should also be attempted to explain how the reader's impression fits into the overall clinical picture. This should be carefully worded so as not to confuse the requesting clinician who may not be a neurologist. Further recommendations as may be relevant, such as longer duration of monitoring or repeating the study after sleep deprivation, can be included here [1].

Sample Normal Report

Routine EEG Report

Name:

Medical Record Number:

Date of Birth:

Reason: This is a 25-year-old man with no significant past medical history presents with episodes of tremulousness.

Relevant medications: none

TECHNICAL SUMMARY: This is a routine EEG performed with anterior temporal as well as standard 10–20 electrodes. Data are recorded using a Nihon Kohden® digital EEG machine.

DETAILS: The waking background shows good organization, consisting of predominantly 30–40 μ V, 9–10 Hz posterior background activity with good reactivity. There is moderate beta activity bilaterally.

Intermittent drowsiness is further characterized by attenuation of the background, slow roving eye movements, and bilateral slowing in the theta and delta range with shifting predominance. Stage N2 sleep is characterized by, further slowing, vertex sharp waves, sleep spindles, K complexes, and posterior occipital sharp transients of sleep (POSTS). Hyperventilation results in mild bilateral buildup. Photoc stimulation results in mild bilateral driving. Single lead EKG shows an apparent sinus regular sinus rhythm.

IMPRESSION: This 30-minute routine EEG with video, performed in the waking, drowsy, and sleeping states is within normal limits. There are no definite epileptiform abnormalities or electrographic seizures during this record. There were no prior EEGs to compare.

CLINICAL CORRELATION: A normal EEG does not exclude the diagnosis of epilepsy.

Sample Abnormal Report

Routine EEG Report

Name:

Medical Record Number:

Date of Birth:

Reason: This is a 67-year-old man with past medical history of hypertension, depression, and peripheral neuropathy who is being evaluated for spells of impaired awareness.

Relevant medications: lisinopril, venlafaxine, cyanocobalamin tablets, and levetiracetam.

Indications: Assess epilepsy risk, interval EEG, now on antiepileptics.

Method: This recording was performed on XLTEK/Natus® equipment. Voltage filters were adjustable, applied as needed after recording. Standard 10–20 EEG electrode placement was used with additional anterior temporal electrodes as needed. The entire EEG dataset was reviewed by eye, with digital spike and seizure detection tools also employed.

DETAILS:

Artifact: Electrode artifact was mild. Myogenic artifact was mild. Artifact did not significantly degrade data quality.

Mental status: Awake and alert.

The waking background was well organized with appropriate posterior dominance and good reactivity to eye closure. The background was symmetric with an amplitude of 20–100 μ V and predominant frequency of 9–10 Hz. Moderate beta activity was present bilaterally.

There were frequent brief bursts (1–5 s) of rhythmic, 50–200 μ V, 2–2.5 Hz slowing seen over the left temporal region (F7, T1, T3). Additionally, there were occasional independent bursts (1–3 s) of higher amplitude (75–300 μ V), polymorphic, bimodal, 1–3 Hz delta slowing.

Drowsiness demonstrated frontocentral slowing, attenuation of the posterior-dominant rhythm and accentuation of above mentioned slowing. Brief sleep demonstrated further slowing, vertex waves, central spindles, and K complexes. Later stages of sleep were not seen.

Hyperventilation was not performed. Photoc stimulation demonstrated moderate driving.

Single EKG channel demonstrated a normal sinus rhythm of 60–80 beats per minute.

Impression: This 30-minute routine EEG with video is abnormal due to the following features:

1. Frequent bursts of left temporal rhythmic slowing consistent with temporal intermittent rhythmic delta activity (TIRDA) which has

a high association with mesial temporal lobe epilepsy (epileptogenic abnormality).

2. Occasional independent bitemporal slowing accentuated during drowsiness and light sleep.

These findings are unchanged in comparison to a prior EEG.

Clinical correlation: This study is consistent with left temporal cortical irritability (seizure tendency) and mild, left greater than right, bitemporal dysfunction. The study captured no events. There were no electrographic seizures during this study.

A Few Sample Interpretations for Reference

“This routine EEG, performed in wakefulness, drowsiness, and slight sleep shows no definite abnormalities. Bilateral irregularities are noted mostly during drowsiness, perhaps right greater than left which do not clearly exceed normal limits for drowsiness. Infrequent wickets are noted and considered a benign pattern of uncertain significance.”

“This routine EEG, performed in the waking, drowsy, and sleep states is abnormal because of bilateral left greater than right hemispheric slowing and rare left sharp waves. This study is consistent with moderate left hemispheric dysfunction and cortical irritability (seizure tendency) in the setting of a mild encephalopathy. A breach rhythm is noted over the left frontotemporal region consistent with his known craniotomy.”

“This portable EEG performed on a patient in the obtunded state is abnormal due to continuous bilateral hemispheric slowing and abundant generalized periodic discharges (GPDs) with a triphasic morphology (also called triphasic waves). This study is consistent with moderate to severe encephalopathy to which medications and/or metabolic dysfunction may contribute.”

“This portable EEG performed on a patient in the confused state reveals

1. Frequent electroclinical seizures with P3 onset, and at times associated with right arm clonic jerking, right gaze deviation, and altered awareness. They occur at a frequency of 8–10 per hour without improvement in the patient’s mental status between seizures, consist with nonconvulsive status epilepticus (NCSE). Seizures stop after intubation at 17:30.
2. Frequent lateralized periodic discharges (LPDs) in the left frontal-central region and isolated discharges over the right parietal region in the initial part of the record.
3. Continuous high-voltage delta slowing over the left side throughout the initial part of record.
4. Predominantly poorly organized, discontinuous, and slow background with burst-suppression of the record after intubation and resolution of NCSE.”

These findings are suggestive of seizures and focal cortical irritability over the left parietal region and left more than right bi-hemispheric cerebral dysfunction. Seizures stop and are replaced by a burst-suppression pattern following intubation and initiation of sedation. The higher voltage on the left side is consistent with a breach artifact due to known skull defect. A prior EEG on -- showed left greater than right hemispheric slowing, with no seizures.”

“This routine EEG performed in the waking and drowsy and states is abnormal because of frequent bursts of generalized epileptiform discharges as described. These findings are indicative of interictal expression of generalized epilepsy syndrome.”

“This routine EEG performed in the waking and excessive drowsy states shows no significant abnormalities. Excessive drowsiness without evidence of sleep may suggest sleep deprivation. An event tremulous involuntary limb and body movements as noted by the technologist during the study is not associated with a definite epileptiform correlate.”

Summary

1. Always treat the patient and not the EEG. Epilepsy is a clinical diagnosis where the EEG is but a single data point.
2. Ideal reports are those that clearly state if the recording is normal or abnormal with appropriate reasoning.
3. Confusing or entirely unexpected findings should arouse suspicion.

Reference

Kaplan PW, SR. Benbadis How to write an EEG report: dos and don'ts. *Neurology*. 2013 Jan 1;80(1 Supp 1): S4 3–6.